Class 12 Mathematics

Chapter 1: Relations and Functions

School Worksheet: step-wise solutions

20 questions **52** marks **3** levels

Each row is one line of working. The red number in the right margin is what an examiner gives for that line, as in the CBSE marking scheme (½ is half a mark). Write every line that carries a mark. ✓ ½

Level 1

8 marks

1 MCQ — 1 mark

Let R be the relation on the set $\{1, 2, 3\}$ given by $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 3)\}$. Then R is:

(A) **reflexive, but neither symmetric nor transitive**

(B) an equivalence relation

(C) symmetric and transitive only

(D) transitive only

Formula / idea: Reflexive: $(a, a) \in R$ for all a. Symmetric: $(a, b) \Rightarrow (b, a)$. Transitive: $(a, b), (b, c) \Rightarrow (a, c)$.

1	Reflexive check: every (a, a) is in R. $(1, 1),\ (2, 2),\ (3, 3) \in R \Rightarrow R$ is reflexive	
2	Symmetric check: $(1, 2)$ is in R but its reverse is not. $(1, 2) \in R$ but $(2, 1) \notin R \Rightarrow$ not symmetric	
3	Transitive check: $(1, 2)$ and $(2, 3)$ are in R, but $(1, 3)$ is not. $(1, 2), (2, 3) \in R$ but $(1, 3) \notin R \Rightarrow$ not transitive	✓ 1
4	So R is reflexive only. Option (A).	

Answer **(A) reflexive, but neither symmetric nor transitive**

Watch out: Calling R transitive without checking the chain $(1, 2), (2, 3) \Rightarrow (1, 3)$.

2 MCQ 1 mark

The number of equivalence relations on the set $\{1, 2, 3\}$ containing $(1, 2)$ and $(2, 1)$ is:

(A) 1 **(B)** 2

(C) 3 **(D)** 4

Formula / idea: An equivalence relation must be reflexive, symmetric and transitive.

1 Smallest one: add all (a, a) to the given pairs.
$R_1 = \{(1,1), (2,2), (3,3), (1,2), (2,1)\}$
R_1 is reflexive, symmetric and transitive.

2 Add one more pair, say $(2, 3)$. Symmetry and transitivity force more pairs.
$(3,2) \in R; \quad (1,2), (2,3) \in R \Rightarrow (1,3) \in R \Rightarrow (3,1) \in R$
Adding $(1, 3)$ or $(3, 1)$ instead leads to the same result.

3 Then every pair is present, so we get $A \times A$.
$R_2 = A \times A$

4 Only R_1 and R_2 are possible. ✓ 1
$\text{Number of equivalence relations} = 2$

Answer **(B)** 2

Watch out: Forgetting that $A \times A$ itself is also an equivalence relation.

3 MCQ 1 mark

The function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x^2$ is:

(A) one-one and onto **(B)** many-one and onto

(C) one-one but not onto **(D)** **neither one-one nor onto**

1 Two different inputs give the same output.
$f(-1) = 1 = f(1), \text{ but } -1 \neq 1 \Rightarrow \text{not one-one}$

2 A negative number has no pre-image, as $x^2 \geq 0$.
$f(x) = -1 \Rightarrow x^2 = -1, \text{ no real } x \Rightarrow \text{not onto}$

3 So f is neither one-one nor onto. Option (D). ✓ 1

Answer **(D) neither one-one nor onto**

4 MCQ 1 mark

The function $f : \mathbb{N} \to \mathbb{N}$ defined by $f(x) = 2x$ is:

(A) bijective

(B) **injective but not surjective**

(C) surjective but not injective

(D) neither injective nor surjective

1 Let $f(x_1) = f(x_2)$.
$2x_1 = 2x_2 \Rightarrow x_1 = x_2 \Rightarrow$ injective (one-one)

2 Try to find a pre-image of $1 \in \mathbb{N}$.
$2x = 1 \Rightarrow x = \frac{1}{2} \notin \mathbb{N} \Rightarrow$ not surjective (onto)

3 So f is injective but not surjective. Option (B). ✓ 1

Answer **(B) injective but not surjective**

Watch out: Taking $x = \frac{1}{2}$ as a pre-image — the domain is $\mathbb{N}$, not $\mathbb{R}$.

5 MCQ 1 mark

If set A has 3 elements and set B has 4 elements, then the number of one-one functions from A to B is:

(A) 12

(B) 24

(C) 64

(D) 81

Formula / idea: Number of one-one functions from an m-set to an n-set ($m \le n$) is ${}^{n}P_{m}$.

1 Images must all be different: 4 choices, then 3, then 2.
${}^{4}P_{3} = 4 \times 3 \times 2 = 24$ ✓ 1

2 Option (B).

Answer **(B)** 24

Watch out: Using $4^3 = 64$, which counts all functions, not just one-one ones.

6 **Assertion-Reason** 1 mark

Assertion (A): The relation $R = \{(a, b) : a \leq b\}$ on the set of real numbers is not an equivalence relation. **Reason (R):** The relation $R = \{(a, b) : a \leq b\}$ is not symmetric.

(A) **Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).**

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

1 Assertion: $2 \leq 3$ but $3 \not\leq 2$, so R fails symmetry. A is true.
$(2, 3) \in R,\ (3, 2) \notin R$

2 Reason: the same example shows R is not symmetric. R is true.

3 R is reflexive ($a \leq a$) and transitive; only symmetry fails. So R explains A.
$a \leq b,\ b \leq c \Rightarrow a \leq c$

4 Option (A). ✓ 1

Answer **(A) Both A and R are true, and R is the correct explanation of A**

7 **Short answer** 2 marks

Show that the function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = 3 - 4x$ is a bijection.

Formula / idea: Bijection = one-one and onto.

1 **One-one:** let $x_1, x_2 \in \mathbb{R}$ with $f(x_1) = f(x_2)$.
$3 - 4x_1 = 3 - 4x_2$

2 Subtract 3 and divide by -4. ✓ 1
$x_1 = x_2 \Rightarrow f$ is one-one

3 **Onto:** let $y \in \mathbb{R}$. Put $f(x) = y$ and solve for x.
$3 - 4x = y \Rightarrow x = \dfrac{3 - y}{4} \in \mathbb{R}$

4 Check that this x maps to y. ✓ 1
$f\left(\dfrac{3 - y}{4}\right) = 3 - 4 \cdot \dfrac{3 - y}{4} = 3 - (3 - y) = y \Rightarrow f$ is onto

5	Hence f is one-one and onto, i.e. a bijection.	

Answer f **is one-one and onto, so** f **is a bijection.**

Watch out: Finding $x = \dfrac{3-y}{4}$ but not checking $f(x) = y$, or not saying $x \in \mathbb{R}$.

Level 2

19 marks

8 Short answer

2 marks

Show that the relation R on the set of integers $\mathbb{Z}$ defined by $R = \{(a, b) : 2 \text{ divides } (a - b)\}$ is an equivalence relation.

1	**Reflexive:** for any $a \in \mathbb{Z}$, $a - a = 0 = 2 \times 0 \Rightarrow 2 \mid (a - a) \Rightarrow (a, a) \in R$	✓ ½
2	**Symmetric:** let $(a, b) \in R$, so $a - b = 2k$, $k \in \mathbb{Z}$. $b - a = -2k = 2(-k) \Rightarrow 2 \mid (b - a) \Rightarrow (b, a) \in R$	✓ ½
3	**Transitive:** let $(a, b), (b, c) \in R$: $a - b = 2k$, $b - c = 2m$. Add. $a - c = 2k + 2m = 2(k + m) \Rightarrow (a, c) \in R$	✓ ½
4	Hence R is an equivalence relation.	✓ ½

Answer R **is reflexive, symmetric and transitive, hence an equivalence relation.**

9 Short answer

3 marks

Check whether the relation R on $\mathbb{R}$ defined by $R = \{(a, b) : a \leq b^3\}$ is reflexive, symmetric or transitive.

Formula / idea: To disprove a property, one counter-example is enough.

1	**Reflexive?** Take $a = \frac{1}{2}$. $\left(\frac{1}{2}\right)^3 = \frac{1}{8}$ and $\frac{1}{2} \not\leq \frac{1}{8} \Rightarrow \left(\frac{1}{2}, \frac{1}{2}\right) \notin R$	
2	So R is not reflexive.	✓ 1
3	**Symmetric?** $1 \leq 2^3 = 8$, so $(1, 2) \in R$. But $2 \not\leq 1^3 = 1 \Rightarrow (2, 1) \notin R$	
4	So R is not symmetric.	✓ 1
5	**Transitive?** Check $\left(3, \frac{3}{2}\right)$ and $\left(\frac{3}{2}, \frac{4}{3}\right)$. $3 \leq \left(\frac{3}{2}\right)^3 = \frac{27}{8}, \quad \frac{3}{2} \leq \left(\frac{4}{3}\right)^3 = \frac{64}{27}$	

6	Both are in R, but $\left(3, \frac{4}{3}\right)$ is not. $3 \not\leq \left(\frac{4}{3}\right)^3 = \frac{64}{27} \approx 2.37 \Rightarrow \left(3, \frac{4}{3}\right) \notin R$	
7	So R is not transitive.	✓ 1

Answer R **is neither reflexive, nor symmetric, nor transitive.**

Watch out: Testing only whole numbers like $a = 1$ or 2 — they satisfy $a \leq a^3$, so pick $a = \frac{1}{2}$.

10 **Short answer** 3 marks

Let $A = \{1, 2, 3, \ldots, 9\}$ and R be the relation on $A \times A$ defined by $(a, b)\, R\, (c, d)$ if $a + d = b + c$. Prove that R is an equivalence relation and find the equivalence class $[(2, 5)]$.

1	**Reflexive:** for any $(a, b) \in A \times A$, $a + b = b + a \Rightarrow (a, b)\, R\, (a, b)$	✓ ½
2	**Symmetric:** let $(a, b)\, R\, (c, d)$. $a + d = b + c \Rightarrow c + b = d + a \Rightarrow (c, d)\, R\, (a, b)$	✓ ½
3	**Transitive:** let $(a, b)\, R\, (c, d)$ and $(c, d)\, R\, (e, f)$. Add the two equations. $a + d = b + c$ $c + f = d + e$ $\Rightarrow a + d + c + f = b + c + d + e$	
4	Cancel $c + d$ from both sides. $a + f = b + e \Rightarrow (a, b)\, R\, (e, f)$	✓ 1
5	Hence R is an equivalence relation.	
6	**Class of** $(2, 5)$**:** $(a, b)\, R\, (2, 5)$ means $a + 5 = b + 2 \Rightarrow b = a + 3$	✓ ½
7	Need $b \leq 9$, so $a = 1, 2, \ldots, 6$. $[(2, 5)] = \{(1, 4), (2, 5), (3, 6), (4, 7), (5, 8), (6, 9)\}$	✓ ½

Answer R **is an equivalence relation;** $[(2, 5)] = \{(1, 4), (2, 5), (3, 6), (4, 7), (5, 8), (6, 9)\}$**.**

Watch out: Including $(7, 10)$ — $10 \notin A$.

11 **Short answer** 3 marks

Let $f : \mathbb{N} \to \mathbb{N}$ be defined by $f(n) = \begin{cases} \dfrac{n+1}{2}, & n \text{ odd} \\ \dfrac{n}{2}, & n \text{ even} \end{cases}$. Show that f is onto but not one-one.

1	**Not one-one:** find $f(1)$ and $f(2)$. $f(1) = \dfrac{1+1}{2} = 1, \qquad f(2) = \dfrac{2}{2} = 1$	
2	$1 \neq 2$ but $f(1) = f(2)$, so f is not one-one.	✓ 1
3	**Onto:** let $m \in \mathbb{N}$. Take the odd number $n = 2m - 1 \in \mathbb{N}$. $f(2m-1) = \dfrac{(2m-1)+1}{2} = m$	✓ 1
4	Also take the even number $n = 2m \in \mathbb{N}$. $f(2m) = \dfrac{2m}{2} = m$	✓ ½
5	Every $m \in \mathbb{N}$ has a pre-image, so f is onto.	✓ ½

Answer f **is onto but not one-one.**

Watch out: Proving onto with one example value of m instead of a general $m \in \mathbb{N}$.

12 **Short answer** 3 marks

Show that the function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = \dfrac{x}{x^2+1}$ is neither one-one nor onto.

1	**Not one-one:** find $f(2)$ and $f\left(\frac{1}{2}\right)$. $f(2) = \dfrac{2}{4+1} = \dfrac{2}{5}, \qquad f\left(\frac{1}{2}\right) = \dfrac{\frac{1}{2}}{\frac{1}{4}+1} = \dfrac{\frac{1}{2}}{\frac{5}{4}} = \dfrac{2}{5}$	✓ ½
2	$2 \neq \frac{1}{2}$ but $f(2) = f\left(\frac{1}{2}\right)$, so f is not one-one.	✓ 1
3	**Not onto:** suppose $1 \in \mathbb{R}$ has a pre-image x. $\dfrac{x}{x^2+1} = 1 \Rightarrow x^2 - x + 1 = 0$	✓ ½
4	Find the discriminant. $D = (-1)^2 - 4(1)(1) = -3 < 0 \Rightarrow$ no real x	✓ ½
5	So 1 has no pre-image; f is not onto.	✓ ½

Answer f **is neither one-one nor onto.**

Watch out: Saying "not onto" without naming a codomain value that has no pre-image.

13 **Short answer** 3 marks

Let $A = \mathbb{R} - \{3\}$ and $B = \mathbb{R} - \{1\}$. Show that $f : A \to B$ defined by $f(x) = \dfrac{x-2}{x-3}$ is one-one and onto.

1	**One-one:** let $x_1, x_2 \in A$ with $f(x_1) = f(x_2)$. Cross-multiply. $(x_1 - 2)(x_2 - 3) = (x_2 - 2)(x_1 - 3)$	✓ ½
2	Expand both sides. $x_1x_2 - 3x_1 - 2x_2 + 6 = x_1x_2 - 3x_2 - 2x_1 + 6$	
3	Cancel equal terms. $-3x_1 - 2x_2 = -3x_2 - 2x_1 \Rightarrow -x_1 = -x_2 \Rightarrow x_1 = x_2$	
4	So f is one-one.	✓ 1
5	**Onto:** let $y \in B$, so $y \neq 1$. Put $f(x) = y$. $x - 2 = xy - 3y \Rightarrow x(1-y) = 2 - 3y \Rightarrow x = \dfrac{3y-2}{y-1}$	✓ ½
6	Check $x \neq 3$: if $x = 3$ then $3y - 2 = 3y - 3 \Rightarrow -2 = -3$, impossible $\Rightarrow x \in A$ So the pre-image lies in the domain A.	✓ ½
7	Check that $f(x) = y$. $f(x) = \dfrac{\frac{3y-2}{y-1} - 2}{\frac{3y-2}{y-1} - 3} = \dfrac{3y - 2 - 2y + 2}{3y - 2 - 3y + 3} = \dfrac{y}{1} = y$	
8	Every $y \in B$ has a pre-image in A, so f is onto.	✓ ½

Answer f **is one-one and onto.**

Watch out: Not checking that $x = \dfrac{3y-2}{y-1}$ is never 3, i.e. that it lies in A.

14 **Short answer** 2 marks

Let $A = \{1, 2, 3, 4, 5, 6\}$ and $R = \{(a, b) : b = a + 1\}$ on A. Is R reflexive, symmetric or transitive? Justify.

1	Write R in roster form. $R = \{(1,2), (2,3), (3,4), (4,5), (5,6)\}$	✓ ½
2	**Reflexive?** $1 \neq 1 + 1$. $(1,1) \notin R \Rightarrow$ not reflexive	✓ ½
3	**Symmetric?** $(1,2) \in R$ but $(2,1) \notin R \Rightarrow$ not symmetric	✓ ½

4 **Transitive?**
$(1,2),(2,3) \in R$ but $(1,3) \notin R \Rightarrow$ not transitive ✓ ½

5 So R is neither reflexive, nor symmetric, nor transitive.

Answer R **is neither reflexive, nor symmetric, nor transitive.**

Level 3

25 marks

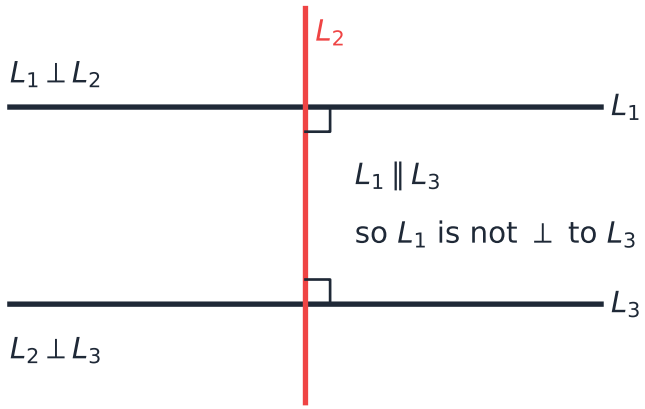
15 **Short answer** 3 marks

Let L be the set of all lines in a plane and $R = \{(L_1, L_2) : L_1 \perp L_2\}$. Show that R is symmetric but neither reflexive nor transitive.

1 **Reflexive?** No line is perpendicular to itself. ✓ 1
$L_1 \not\perp L_1 \Rightarrow (L_1, L_1) \notin R \Rightarrow$ not reflexive

2 **Symmetric:** let $(L_1, L_2) \in R$. ✓ 1
$L_1 \perp L_2 \Rightarrow L_2 \perp L_1 \Rightarrow (L_2, L_1) \in R$

3 **Transitive?** Let $L_1 \perp L_2$ and $L_2 \perp L_3$.
$L_1 \perp L_2,\ L_2 \perp L_3 \Rightarrow L_1 \parallel L_3$
Two lines perpendicular to the same line are parallel.



4 So $L_1 \not\perp L_3$, i.e. $(L_1, L_3) \notin R$. Not transitive. ✓ 1

5 Hence R is symmetric but neither reflexive nor transitive.

Answer R **is symmetric but neither reflexive nor transitive.**

16 **Long answer** 5 marks

Let $\mathbb{N}$ be the set of natural numbers and R be the relation on $\mathbb{N} \times \mathbb{N}$ defined by $(a, b)\, R\, (c, d)$ if and only if $ad = bc$. Show that R is an equivalence relation.

1	**Reflexive:** for any $(a, b) \in \mathbb{N} \times \mathbb{N}$, $ab = ba \Rightarrow (a, b)\, R\, (a, b)$ Multiplication of natural numbers is commutative.	✓ 1
2	**Symmetric:** let $(a, b)\, R\, (c, d)$. $ad = bc \Rightarrow cb = da \Rightarrow (c, d)\, R\, (a, b)$	✓ 1
3	**Transitive:** let $(a, b)\, R\, (c, d)$ and $(c, d)\, R\, (e, f)$. $ad = bc \quad \text{and} \quad cf = de$	✓ ½
4	Multiply the two equations. $(ad)(cf) = (bc)(de)$	✓ ½
5	Divide both sides by cd. $af = be$ $c, d \in \mathbb{N}$, so $cd \neq 0$.	✓ 1
6	So $(a, b)\, R\, (e, f)$; R is transitive.	✓ ½
7	Hence R is an equivalence relation.	✓ ½

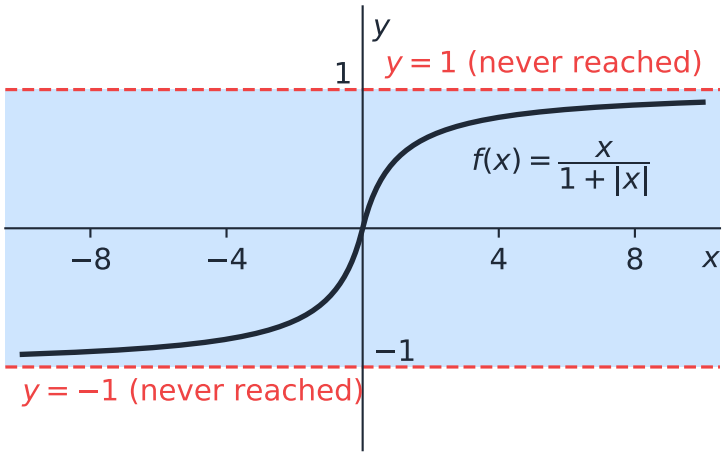
Answer R **is reflexive, symmetric and transitive, hence an equivalence relation.**

Watch out: Dividing by cd without saying $cd \neq 0$ because $c, d \in \mathbb{N}$.

17 **Long answer** 5 marks

Show that the function $f : \mathbb{R} \to \{x \in \mathbb{R} : -1 < x < 1\}$ defined by $f(x) = \dfrac{x}{1+|x|}$ is one-one and onto.

Formula / idea: Split $|x|$: $|x| = x$ for $x \geq 0$ and $|x| = -x$ for $x < 0$.



1	Write f without the modulus. $f(x) = \begin{cases} \dfrac{x}{1+x}, & x \geq 0 \\ \dfrac{x}{1-x}, & x < 0 \end{cases}$	✓ ½
2	**One-one, Case 1:** $x_1, x_2 \geq 0$ and $f(x_1) = f(x_2)$. Cross-multiply. $x_1 + x_1x_2 = x_2 + x_1x_2 \Rightarrow x_1 = x_2$	✓ ½
3	**Case 2:** $x_1, x_2 < 0$ and $f(x_1) = f(x_2)$. Cross-multiply. $x_1 - x_1x_2 = x_2 - x_1x_2 \Rightarrow x_1 = x_2$	✓ ½
4	**Case 3:** $x_1 \geq 0, x_2 < 0$. Signs of the outputs differ. $f(x_1) \geq 0 > f(x_2) \Rightarrow f(x_1) \neq f(x_2)$	✓ ½
5	In all cases $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$. So f is one-one.	✓ ½
6	**Onto, Case 1:** let $0 \leq y < 1$. Take $x = \dfrac{y}{1-y} \geq 0$. $f(x) = \dfrac{\frac{y}{1-y}}{1+\frac{y}{1-y}} = \dfrac{\frac{y}{1-y}}{\frac{1}{1-y}} = y$ $1 - y > 0$, so $x \geq 0$.	✓ 1
7	**Case 2:** let $-1 < y < 0$. Take $x = \dfrac{y}{1+y} < 0$. $f(x) = \dfrac{\frac{y}{1+y}}{1-\frac{y}{1+y}} = \dfrac{\frac{y}{1+y}}{\frac{1}{1+y}} = y$ $1 + y > 0$ and $y < 0$, so $x < 0$.	✓ 1
8	Every $y \in (-1, 1)$ has a pre-image, so f is onto.	✓ ½

9	Hence f is one-one and onto.	

Answer f **is one-one and onto.**

Watch out: Using one formula for all x — the pre-image formula differs for $y \geq 0$ and $y < 0$.

18 **Long answer** 5 marks

Let $A = \{x \in \mathbb{Z} : 0 \leq x \leq 12\}$. Show that $R = \{(a, b) : |a - b| \text{ is a multiple of } 4\}$ is an equivalence relation on A. Find the set of all elements related to 1 and write the equivalence class $[2]$.

1	**Reflexive:** for any $a \in A$, $\lvert a - a \rvert = 0 = 4 \times 0 \Rightarrow (a, a) \in R$	✓ 1
2	**Symmetric:** let $(a, b) \in R$, so $\lvert a - b \rvert = 4k$. $\lvert b - a \rvert = \lvert a - b \rvert = 4k \Rightarrow (b, a) \in R$	✓ 1
3	**Transitive:** let $(a, b), (b, c) \in R$: $a - b = \pm 4k$, $b - c = \pm 4m$. $a - c = (a - b) + (b - c) = \pm 4k \pm 4m = 4(\pm k \pm m)$	
4	So $\lvert a - c \rvert$ is a multiple of 4; $(a, c) \in R$.	✓ 1
5	Hence R is an equivalence relation on A.	
6	**Related to 1:** need $\lvert 1 - b \rvert \in \{0, 4, 8, \ldots\}$ with $b \in A$. $b = 1,\ 5,\ 9 \Rightarrow \{1, 5, 9\}$	✓ 1
7	**Class $[2]$:** need $\lvert 2 - b \rvert$ a multiple of 4, $b \in A$. $b = 2,\ 6,\ 10 \Rightarrow [2] = \{2, 6, 10\}$	✓ 1

Answer R **is an equivalence relation; elements related to** 1**:** $\{1, 5, 9\}$**;** $[2] = \{2, 6, 10\}$**.**

Watch out: Including 13 or -3 — they are not in $A = \{0, 1, \ldots, 12\}$.

19 **Short answer** 3 marks

Show that the function $f : [-1, 1] \to \mathbb{R}$ defined by $f(x) = \dfrac{x}{x + 2}$ is one-one. Also find the range of f.

1	**One-one:** let $x_1, x_2 \in [-1, 1]$ with $f(x_1) = f(x_2)$. Cross-multiply. $x_1(x_2 + 2) = x_2(x_1 + 2)$	✓ ½
2	Expand and cancel $x_1 x_2$. $2x_1 = 2x_2 \Rightarrow x_1 = x_2 \Rightarrow f$ is one-one	✓ 1
3	**Range:** rewrite $f(x)$. $f(x) = \dfrac{(x + 2) - 2}{x + 2} = 1 - \dfrac{2}{x + 2}$	

4	For $-1 \le x \le 1$, we get $1 \le x+2 \le 3$. $\frac{2}{3} \le \frac{2}{x+2} \le 2 \Rightarrow -1 \le f(x) \le \frac{1}{3}$	✓ ½
5	Conversely, let $-1 \le y \le \frac{1}{3}$. Solve $y = \frac{x}{x+2}$ for x. $x = \frac{2y}{1-y}$	✓ ½
6	As $1-y>0$: $y \ge -1 \Rightarrow x \ge -1$ and $y \le \frac{1}{3} \Rightarrow x \le 1$. $x \in [-1,1]$ and $f(x) = y$	
7	So every such y is attained. Range of $f = \left[-1, \frac{1}{3}\right]$ Ends: $f(-1) = -1$, $f(1) = \frac{1}{3}$.	✓ ½

Answer f **is one-one; range of** $f = \left[-1, \frac{1}{3}\right]$.

Watch out: Giving the range as $\mathbb{R} - \{1\}$ — that ignores the domain $[-1, 1]$.

20 **Case study** 4 marks

Case Study: A classroom teacher wants to assess her students' understanding of "relations". She writes the following five relations, each defined on the set $A = \{1, 2, 3\}$: $R_1 = \{(2,3), (3,2)\}$, $R_2 = \{(1,2), (1,3), (3,2)\}$, $R_3 = \{(1,2), (2,1), (1,1)\}$, $R_4 = \{(1,1), (1,2), (3,3), (2,2)\}$, $R_5 = \{(1,1), (1,2), (3,3), (2,2), (2,1), (2,3), (3,2)\}$.

(i) Identify the relation which is reflexive, transitive but not symmetric. 1 mark

(ii) Identify the relation which is reflexive and symmetric but not transitive. 1 mark

(iii)(a) Identify the relations which are symmetric but neither reflexive nor transitive. 2 marks

OR

(iii)(b) What pairs should be added to the relation R_2 to make it an equivalence relation? 2 marks

Formula / idea: Reflexive on A needs all of $(1,1), (2,2), (3,3)$.

(i) 1 mark

1	Only R_4 and R_5 contain $(1,1), (2,2), (3,3)$; R_5 is symmetric. Test R_4.	
2	R_4 is reflexive; $(1,2) \in R_4$ but $(2,1) \notin R_4$, so not symmetric.	
3	Transitive: the only chains through $(1,2)$ give $(1,2)$ again. $(1,1), (1,2) \Rightarrow (1,2) \in R_4$; $(1,2), (2,2) \Rightarrow (1,2) \in R_4$	
4	So the required relation is R_4.	✓ 1

Answer (i) R_4

(ii) 1 mark

1	R_5 contains $(1,1),(2,2),(3,3)$, so it is reflexive.	
2	Symmetric: $(1,2),(2,1) \in R_5$ and $(2,3),(3,2) \in R_5$.	
3	Not transitive: $(1,2),(2,3) \in R_5$ but $(1,3) \notin R_5$	
4	So the required relation is R_5.	✓ 1

Answer (ii) R_5

(iii)(a) 2 marks

1	R_1: $(1,1) \notin R_1$ (not reflexive); $(2,3),(3,2)$ both present (symmetric). $(2,3),(3,2) \in R_1$ but $(2,2) \notin R_1 \Rightarrow$ not transitive	✓ 1
2	R_3: $(2,2) \notin R_3$ (not reflexive); $(1,2),(2,1)$ both present (symmetric). $(2,1),(1,2) \in R_3$ but $(2,2) \notin R_3 \Rightarrow$ not transitive	✓ 1
3	R_2 is not symmetric; R_4, R_5 are reflexive. So the answer is R_1 and R_3.	

Answer (iii)(a) R_1 **and** R_3

OR

(iii)(b) 2 marks

1	For reflexive, add the missing diagonal pairs. $(1,1),\ (2,2),\ (3,3)$	✓ ½
2	For symmetric, add the reverse of each pair of R_2. $(2,1),\ (3,1),\ (2,3)$	✓ ½
3	The new relation has all 9 pairs, so it is $A \times A$. $R_2 \cup \{\ldots\} = A \times A$ $A \times A$ is reflexive, symmetric and transitive.	✓ ½
4	Pairs to add: $(1,1),(2,2),(3,3),(2,1),(3,1),(2,3)$.	✓ ½

Answer (iii)(b) **Add** $(1,1),(2,2),(3,3),(2,1),(3,1),(2,3)$**; then** R_2 **becomes** $A \times A$**.**

Watch out: Calling R_1 transitive — $(2,3),(3,2)$ forces $(2,2)$, which is missing.